

CSCE 679 DATA VISUALIZATION · MODULE 03 · LECTURE NOTES

Data Types and Marks

What the data is determines what the channel can be.

MODULE OUTCOMES

- Classify attributes by measurement type.
- Explain which channels each type permits.
- Explain keys against values and why it matters.
- Explain derived attributes and when to compute them.
- Diagnose a type-channel mismatch.

1 Measurement types

Type	What it permits	Example
Nominal	Equality comparison only.	Country, product category, error code.
Ordinal	Equality and order, but not distance.	Small / medium / large; survey agreement levels; severity.
Interval	Order and differences, with no meaningful zero.	Temperature in Celsius; calendar dates.
Ratio	All of the above, plus meaningful ratios.	Count, mass, revenue, duration, distance.

The interval/ratio distinction decides whether a ratio statement is meaningful — 20 degrees Celsius is not twice 10 degrees, because the zero is arbitrary — and that is also exactly why an area or length encoding of an interval variable is nonsense (§1's callout). The Celsius example settles the question quickly and is worth keeping to hand, because the distinction otherwise feels pedantic until it produces a wrong chart.

THE TYPE DETERMINES WHICH CHANNELS ARE EVEN AVAILABLE

A nominal attribute cannot go on a quantitative channel — putting country codes on a numeric axis implies both an order and a spacing that do not exist, and readers will read both of them as meaningful (Module 02 §3's attributed-meaning point).

An ordinal attribute can go on an ordered channel and must not go on a continuous one — the spacing between 'small' and 'medium' is undefined, so a position encoding on a continuous axis asserts something the data does not contain.

And a ratio attribute is the only type for which length and area are meaningful, because both of those channels are read as ratios from a zero (Module 02 §1's shared-reference argument) — so the zero has to mean something.

Which gives a clean diagnostic: a bar chart of an interval variable is a type error — a bar's length asserts a ratio from zero, and an interval scale has no meaningful zero, so the figure makes a claim the data cannot support. A line or a point is correct there instead.

2 Keys and values

A KEY IDENTIFIES AN ITEM; A VALUE IS MEASURED ABOUT IT

A key is independent — it is the thing you are looking at, and usually what you are indexing, grouping, or faceting by. **A value is dependent** — it is the measurement made about the item the key identifies.

And the chart's structure follows directly: keys typically go on spatial position or on faceting, and values go on the quantitative channels — which is why most charts look the way they do, and makes the convention intelligible rather than arbitrary.

With the number of keys determining the chart family: one key gives a bar chart or a line; two keys give a heatmap or a grouped chart; three or more require a faceted arrangement (Module 06 §1).

So counting the keys and the values is the first step in choosing an encoding — and it is considerably more informative than knowing the names of chart types, because it narrows the available choices before any aesthetic question arises.

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1 KEY, 1 VALUE
  bar chart (key on position, value on length)
  or a line if the key is ordered and continuous

1 KEY, SEVERAL VALUES
  grouped or stacked bars; a parallel coordinates
  plot; a radar chart (and radar charts inherit the
  angle problem of Module 02 section 1)

2 KEYS, 1 VALUE
  a heatmap (both keys on position, the value on
  colour) or a grouped bar chart

0 KEYS, 2 VALUES
  a scatter plot -- both values on position, which
  is why it is the most accurate chart there is
  (Module 02 section 1)

0 KEYS, 3+ VALUES
  a scatter plot matrix, or dimensionality
  reduction first (CSCE 676 Module 05)

S0: count first, then choose.

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A scatter plot puts both values on position — rank one, twice — **which is why it is the most accurate chart available and should be the default for two quantities. The scatter-plot-is-rank-one-twice observation is worth making explicit**, because it explains why scatter plots are so effective and why replacing one with a bar chart of binned values throws away accuracy (Module 05 §3's binning point).

3 Derived attributes

- **If the reader's task is a difference, plot the difference. Two lines and a mental subtraction is far worse than one line showing the difference directly — because subtraction is not a visual channel** and the reader will estimate it badly, especially where the lines are far apart.
- **If the task is a ratio or a rate, compute it** — per capita, per unit time, per unit area, or as a proportion of a total — rather than showing the numerator and the denominator and expecting division.
- **If the task is a change, plot the change**, not two snapshots — **which is the commonest missed derivation**, and which turns a comparison task into a lookup task.
- **And if the task is a rank, plot the rank** rather than the value, where the ordering is what matters and the magnitudes do not.
- **Which is a restatement of Module 01 §2: the reader's task determines the data you should be plotting, not only the encoding you put it on** — and that is the broader and more useful version of the principle. **Plot the difference rather than two lines** is the one-line summary.

AND NORMALISATION IS A DERIVATION WITH A LARGE EFFECT ON THE CONCLUSION

Raw counts, per capita figures, per unit area, and year-on-year change can each produce a different and individually defensible picture of exactly the same data — and a map of raw counts is largely a map of population, which is Module 08 §3's central warning.

So the normalisation is an analytical choice that has to be stated, not a formatting step — and the reader cannot infer which one you used from the figure unless the axis label says so explicitly.

And the wrong normalisation is a common honest error: dividing by the wrong denominator produces a confident and wrong conclusion that nothing in the figure contradicts — which Module 08 §3 treats for geographic data where it is most frequent.

Which is why the axis label is load-bearing rather than decorative — 'cases' and 'cases per 100,000' are different claims, and a figure that does not say which it is showing is not interpretable at all, however well encoded.

4 Diagnosing mismatches

- **Nominal on a position axis without a stated order** — which **implies an order the data does not have**, and readers will read that order as meaningful (Module 02 §3). See the fix below.
- **Ordinal spaced as though it were continuous** — equal gaps between 'rarely', 'sometimes', and 'often' — which asserts distances that are undefined and invites arithmetic on them.
- **Interval encoded on length or area** — which asserts a ratio from a zero that does not exist (§1's diagnostic), and is why temperature bar charts are wrong and temperature lines are right.
- **Quantitative on colour hue** — which is **Module 02 §1's rank seven** and is **Module 04's main subject**, and is probably the single most common mismatch in published figures.

- **And a continuous variable binned without saying so** — where **the bin boundaries then do work the data did not**, and a different binning gives a different picture (Module 05 §3). **Sorting a nominal axis by value is the fix for the first row: it gives the reader a meaningful order instead of an arbitrary one, and it costs nothing.**

AND SORTING IS THE CHEAPEST IMPROVEMENT AVAILABLE

A bar chart with its categories in alphabetical order, or in whatever order the source file had, makes the reader search for everything — and the same chart sorted by value makes the ranking, the extremes, and the shape of the distribution readable at a glance.

Which costs one line of code and improves every reader task except looking up a specific named category — and that task is better served by a table in any case (Module 01 §3's division of labour).

So: sort by value unless there is a meaningful order — and if there is a meaningful order, respect it, which is the ordinal case from §1 and is why sorting a month axis by value would be wrong.

And it is a good illustration of this module's whole point: the data's type determined that sorting was even available (nominal, so no inherent order to violate), and the reader's task determined that it was worth doing — which is §1 and Module 01 §2 working together.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Munzner, chapters 2 and 3 (slides free)	§1 and §2 — the attribute types, the key/value distinction, and the dataset taxonomy.
Stevens — On the theory of scales of measurement (free)	§1's four types in the original — short, and the permitted-operations framing comes from here.
Wilkinson — The Grammar of Graphics	§2 and §3 formalised — the grammar that ggplot2 and Vega-Lite implement. Library copy.
Wilke, the chapters on directory of visualizations (free)	§2's counting rules applied — organised by what the data looks like rather than by chart name.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Classify twenty attributes** from a real dataset by measurement type.

- 2 **Find an interval variable** and explain why a bar chart of it is wrong.
- 3 **Find a nominal variable on a numeric axis** in a published figure.
- 4 **Count the keys and values** for five datasets, and predict the chart family.
- 5 **Plot two series and then their difference**, and ask five people to estimate the gap from each.
- 6 **Compute a rate from two raw counts** and compare the two figures.
- 7 **Normalise the same data three ways** and write the three different conclusions.
- 8 **Find a figure whose axis label does not say the normalisation.**
- 9 **Sort an alphabetical bar chart by value** and compare the readability.
- 10 **Find a case where sorting would be wrong**, and say why.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the four measurement types and what each permits.
- 2 Why is 20 degrees Celsius not twice 10, and what follows for encoding?
- 3 Why can nominal data not go on a numeric axis?
- 4 Why is a bar chart of an interval variable a type error?
- 5 Distinguish a key from a value, and say where each goes.
- 6 Give the chart family for four key/value counts.
- 7 Why is a scatter plot the most accurate chart?
- 8 Give four derivations and the commonest missed one.
- 9 Why is normalisation a reportable decision?
- 10 Give five type-channel mismatches, and the cheapest improvement.